

SSA -> Best-in-Class Sovereign Sailing Performance Analyst -- Architecture & Roadmap

Unified plan. Reconciles the debrief pipeline on `main`, the query/eval work on branch `ai-sovereign-mistral`, the earlier Cowork handover (`SSA_AI_STACK_HANOVER.md`), and 2026 best-in-class research. Date: 2026-07-18.

0. Where we actually are (reconciliation)

Two complementary workstreams exist, on two branches:

Workstream	Branch	Files	State
Debrief transcription + summary	<code>main</code>	<code>api/ai/transcribe</code> (whisper-large-v3), <code>api/ai/debrief-summary</code> (mistral-small, single-shot), <code>lib/debriefAudio.js</code> , <code>lib/debriefGlossary.ts</code>	Working; single-shot summary is the weak point
NL query + feedback/eval loop	<code>ai-sovereign-mistral</code> (pushed, not merged)	<code>api/ai/analyze</code> , <code>api/ai/feedback</code> , <code>api/ai/library-search</code> , <code>lib/ai/mistral.ts</code> , <code>scripts/ai-eval.mjs</code> , <code>evals/</code> , <code>ai_query_log</code>	Working but naive (no router/tools/RAG)

Two immediate hazards: 1. **Migration collision** -- both branches ship `0055_*`. `main` = `0055_windweight_forecast.sql`, branch = `0055_ai_query_log.sql`. Renumber the branch's to `0056_ai_query_log.sql` before merge. 2. **forecast-summary uses Anthropic** -- the one non-sovereign exception. Leave it, but it must **never** be fed debrief content. Flagged.

Hard constraint (binding): all inference stays inside the **Scaleway EU** account or **self-hosted on the Mac Studio** (M3 Max, 64 GB). No third-party vendor.

Priorities (next ~2 months): (1) debrief summarisation quality, (2) NL query over SSA data, (3) physics/VPP grounding. IRC deprioritised.

1. What "best-in-class" means in 2026 (research synthesis)

From how SailGP/Oracle, Njord/SailLab, SailSync.ai and Kinetix operate, plus the dbt/Cube semantic-layer benchmarks:

- A performance analyst sits in every debrief** and *simplifies* the data firehose into a few decisions ([SailGP](#)). Our AI plays that role.
- Nobody ships naked text-to-SQL.** Semantic-layer/typed-tool approaches hit 98-100% on covered queries vs 84-90% raw; but the semantic layer scores **0%** off-scope where raw SQL gets 100%. -> **Router:** deterministic typed tools first, sandboxed read-only SQL as the escape hatch, and *tell the user which path it took* ([dbt 2026](#)).
- Expose typed tools, not a schema.** The 24B model never writes the analysis -- it picks a function and fills arguments; Postgres does the maths.
- Hybrid retrieval, reranked, with structured pre-filters** (date/wind-band/boat/sail applied as SQL *before* semantic search) ([Supabase pgvector](#)).
- Evals are a first-class artefact** -- a golden set re-run on every prompt change. The difference between a demo and a tool a coach trusts.

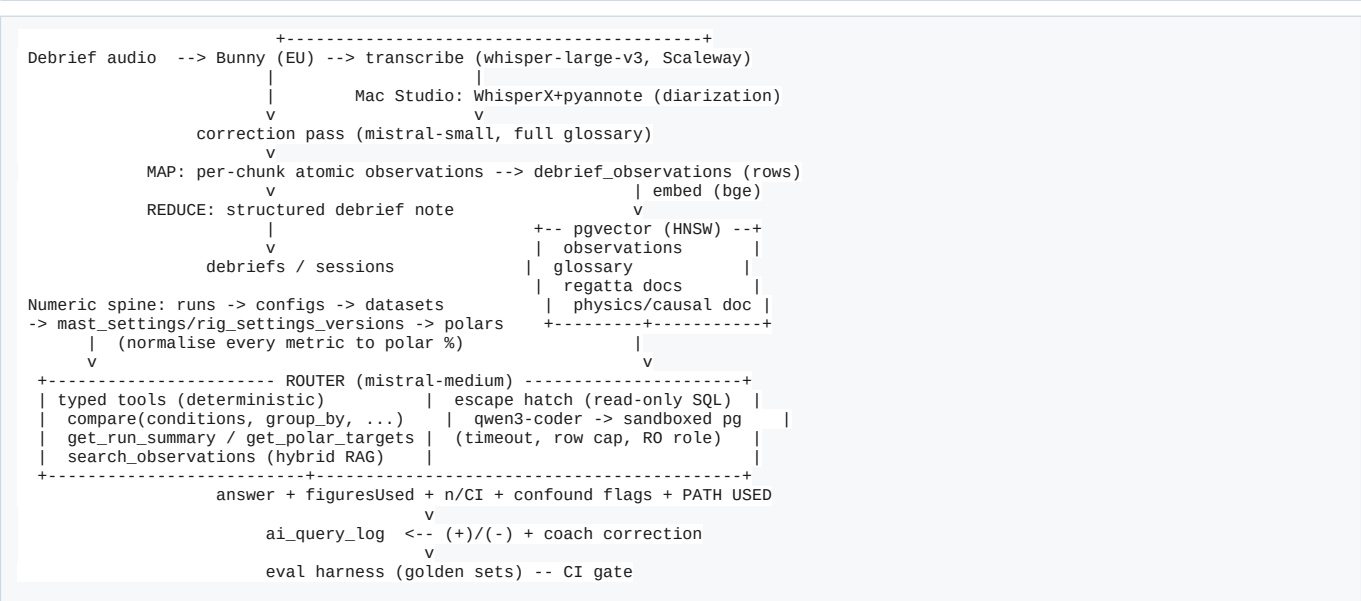
6. **Small model + heavy scaffolding beats big model + naive prompt** -- convenient, since Scaleway constrains model choice anyway.
7. **Debriefs follow a structure** -- a PEARLS-like arc (react -> describe situation -> analyse decisions -> summarise learnings/actions), moderated, blame-free ([elite-sailing debrief study](#)). This is the schema our summariser should target.

2. Sovereign model inventory (provisioned on the [ssa-ai](#) project, verified live)

Role	Model (Scaleway)	Notes
Orchestration / router / narration	mistral-medium-3.5-128b	Frontier reasoning+vision; use for the <i>orchestration path only</i>
Mechanical passes (correction, map-extract, summarise)	mistral-small-3.2-24b-instruct-2506	Current default; cheap
SQL escape-hatch generation	qwen3-coder-30b-a3b-instruct	Code model, for the read-only SQL fallback
Embeddings (RAG)	bge-multilingual-gemma2	Multilingual -> handles Dutch+English debriefs
Transcription	whisper-large-v3	Already wired in transcribe/route.ts
Vision (sail scans, invoices)	mistral-small-3.2-24b / pixtral-12b-2409	--

Not available on Scaleway: no reranker (-> do LLM-rerank or skip), **no Voxtral** (-> diarization must be **WhisperX + pyannote** on the **Mac Studio**, the sovereign route). Also available if needed: [qwen3-235b](#), [qwen3.5-397b](#), [llama-3.3-70b](#), [glm-5.2](#).

3. Target architecture (two goals, one stack)



Everything downstream of RLS (per-team/boat), synchronous Scaleway calls (zero-retention), Mac Studio for diarization only.

4. Data-model additions

Table	Purpose
<code>debrief_observations</code>	Atomic, queryable rows from the map step: <code>{claim, topic, conditions, crew, confidence, ts, session_id, run_id?, embedding}</code> . The bridge between goal 1 and goal 2.
<code>doc_embeddings</code>	pgvector chunks for glossary + regatta docs + physics/causal doc (hybrid: <code>tsvector</code> + <code>vector</code>).
<code>calibration_versions</code>	Versioned calibration record per boat/date -- see the data rule below.
<code>ai_query_log</code>	From the branch (renumber to <code>0056</code>). Unifies feedback across all AI surfaces.

Grain: the unit of performance is the `run` (steady-state segment), not the session. Everything joins `run -> conditions -> sail combo -> mast_settings/rig_settings_versions -> polar target`.

Normaliser (mandatory, first): express every metric as polar %. Without it, wind strength dominates every comparison and the tool learns nothing.

Data-architecture rule (applies to everything): *store raw channels + a versioned calibration record; derive computed quantities at query time*. If any calibration constant changed across the 3 years of Northstar 72 data (recalibration, winter refit, new navigator), the archive is on an inconsistent basis -- every cross-year comparison is comparing different metrics, and anything fitted under an old calibration (polars included) needs refitting. **Audit this before trusting cross-year analysis.**

5. Systemised roadmap (step by step)

Each phase: **Goal · Build · Done-when (eval)**. Order follows the handover, interleaved with the merge + RAG + router dependencies.

Phase 0 -- Reconcile & unify the foundation

- **Build:** Renumber branch migration -> `0056_ai_query_log.sql`; merge `ai-sovereign-mistral` into `main`. Collapse `transcribe`, `debrief-summary`, `analyze` onto the **one** client `lib/ai/mistral.ts` (single model-config, one `json_schema/mistralJSON` helper, GET `/models` health). Keep `forecast-summary` (Anthropic) isolated.
- **Done-when:** both workstreams on `main`; one AI client; `npm run eval:ai` runs; no dup migration numbers.

Phase 1 -- Eval sets (do before touching prompts)

- **Build:** (a) **Debrief eval** -- 10 debriefs, hand-corrected transcripts + ideal notes; score **WER** on transcript, **fact-recall** + **hallucination-rate** on summary. (b) **Query eval** -- ~50 real Q->A, extending `evals/ai-analyze/gold.jsonl`.
- **Done-when:** both baselines recorded; every later change is measured, not vibes.

Phase 2 -- Debrief pipeline v2 (highest leverage)

- **2a Map->reduce** -- per-chunk **atomic structured observations** -> reduce into the note. Kills the 4000-token truncation (delete most of `extractJson`) **and** makes each observation a queryable/embeddable row (`debrief_observations`).

- **2b Correction pass** -- transcribe raw -> dedicated glossary-correction pass with the *full* glossary in `mistral-small` context -> then summarise. `fixups` becomes eval-derived, not hand-maintained.
- **2c Split the one-prompt-four-jobs** -- translate (NL->EN) / repair jargon / structure / judge relevance become separate passes.
- **2d json_schema strict** (if the Scaleway endpoint supports it) -> delete `coerce()` + half of `extractJson()`.
- **2e Audio overlap** -- ~10 s overlap on the 300 s chunks in `debriefAudio.js` + text-level dedup (stop dropping words at hard cuts).
- **2f Diarization** -- WhisperX + pyannote on the **Mac Studio** -> speaker labels; retire the "don't guess who said what" workaround.
- **Done-when:** eval shows $\uparrow$ fact-recall, $\downarrow$ hallucination, zero truncation; observation rows populate.

Phase 3 -- RAG foundation

- **Build:** enable `pgvector`; embed `debrief_observations` + glossary + regatta docs + physics doc with `bge-multilingual-gemma2`; **HNSW** index; **hybrid** search (`tsvector` + vector) with **structured SQL pre-filters** (date/wind-band/boat/sail) applied *before* semantic search.
- **Done-when:** retrieval eval (recall@k) passes on the golden set.

Phase 4 -- The analyst: router + typed tools

- **Build:** a **router** (`mistral-medium`) that prefers deterministic **typed tools**, falls back to **sandboxed read-only SQL** (`qwen3-coder`), and **discloses the path taken**. First tool = `compare(conditions_filter, group_by=setting, metric=polar_pct, min_n)` -> **effect size + confidence interval + n**. Others: `get_run_summary`, `get_polar_targets`, `search_observations` (Phase-3 RAG), `query_datasets`. **Guardrails enforced in tool output** (not the prompt): never report a difference without n + uncertainty; always surface the applied filter; read-only role + statement timeout + row cap. Replace the naive `analyze` route with the router; wire `ai_query_log` + (+)/(-) across.
- **Done-when:** query golden-set pass-rate $\geq$ baseline+; every comparison carries n/CI; path always disclosed.

Phase 5 -- Physics grounding

- **Build:** (1) polar-% normaliser (done in P4) -- *the VPP already doing work*. (2) **Plausibility bounds** -- flag e.g. "12% speed from 2 mm rake -> outside physical plausibility, likely a confound." (3) **Causal-vocabulary doc** as retrieval context (forestay sag -> jib entry -> pointing; rake -> helm balance -> rudder drag; heel -> twin-rudder toe loading). Precompute the polar surface and interpolate -- **do not run a VPP in the query loop**. (4) Execute the **calibration-versioning audit** (§4).
- **Done-when:** plausibility guard catches synthetic implausible claims; calibration basis documented; anything under old calibration flagged for refit.

Phase 6 -- Unify & productionise

- **Build:** one conversational analyst surface (replaces the interim panel) grounded over numeric + observations + docs + fleet; **proactive auto-debrief** per session (transcript + numbers -> draft note). CI eval gate; few-shot curation from (+) rows. **Fine-tune only if the eval plateaus** (it breaks zero-retention -- see [[sovereign-ai-scaleway-mistral]]).
- **Done-when:** KPIs move (below); coach trusts it unprompted.

Phase 7 -- IRC (later; deprioritised)

Rating optimisation (marginal TCC of a bigger A2), competitor corrected-time benchmarking, transfer priors from similar-ratio boats. **Check RORC/UNCL licensing before bulk-ingesting certificate data.**

6. Sovereignty, privacy & guardrails

- **Inference:** Scaleway EU (synchronous -> zero-retention) or Mac Studio. [forecast-summary](#) (Anthropic) is the lone exception -- keep debrief content out of it.
- **GDPR -- voice is personal data.** Recording team debriefs captures identifiable voices -> needs **explicit consent + a retention policy + access control**. Store audio in Bunny (EU); define how long transcripts/recordings live. Decide before scaling Phase 2f.
- **Isolation:** every tool RLS-scoped by team/boat; read-only DB role for the SQL escape hatch.

7. Cost (indicative, per team/month -- trivial)

Transcription [whisper-large-v3](#) on Scaleway + embeddings ([bge](#)) + chat (small/medium) all pay-per-use in the single-digit-euro range at team volume; diarization on the Mac Studio is free compute. Well inside the ~EUR 100/mo envelope.

8. Success metrics

Coach (+)-rate on debrief notes and query answers; eval pass-rate trend; hallucination-rate ↓; time-to-debrief ↓; % answers via deterministic tool vs SQL fallback; adoption per campaign.

9. Open decisions (need your call)

1. **Orchestration model:** [mistral-medium-3.5-128b](#) (recommended) vs [qwen3-235b/qwen3.5-397b](#).
2. **Mac Studio for diarization** (WhisperX+pyannote) -- confirm it's the sovereign home for that.
3. **Merge branch now** (with the [0056](#) renumber) or keep query work isolated until the router lands?
4. **Calibration-versioning audit** -- is there a versioned calibration record across the 3 years, or is cross-year analysis currently unsafe?
5. **Debrief recording consent/retention policy** -- needed before Phase 2 scales.

10. Sources

SailGP analyst workflow ([Yacht Racing Life](#)) · elite-sailing debrief structure ([ScienceDirect](#)) · SailLab/Njord ([sailnjord](#)) · SailSync.ai · Kinetix ([Sailing World](#)) · dbt semantic-layer-vs-text-to-SQL 2026 ([dbt](#)) · Supabase pgvector RAG ([Supabase](#)) · Scaleway supported models ([Scaleway](#)) · plus the earlier [SSA_AI_STACK_HANDOVER.md](#) and [SSA_ML_RESEARCH_REPORT.md](#).